THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER’ S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

MPWAPWA DISTRICT COUNCIL

FORM ONE MID-TERM EXAMINATION

031 PHYSICS

Duration SEPTEMBER 2026

Instructions

1. This paper consists of sections A, B and C with a total of ten (10) questions.
2. Answer all questions in the spaces provided.
3. Section A and C carries fifteen (15) marks each and section B carries seventy (70) marks.
4. All writing must be in blue or black ink.
5. Communication devices and unauthorised materials are not allowed in the examination room.
6. Calculators and mathematical tables may be used in the assessment room.
7. Write your name at the top right corner of every page.
8. The following information may be useful:
   (i) Acceleration due to gravity, g = 10 m/s2.
   (ii) Pie =3.14.

SECTION A (15 Marks)

Answer all questions in this section.

1. For each of the items (i) - (x), choose the correct answer from the given alternatives and write its letter in the box provided. **(10 marks)**

(i) Which of the following is **NOT** a branch of Physics?

A. Mechanics C. Histophysics

B. Light D. Electronics

(ii) Which of the following is the SI unit of volume

A. m^3 C. kilogram

B. Litre D. m^2

(iii) Why does a solid sink in water?

A. Because its volume is large

B. Because its mass is small

C. Because its density is greater than that of water

D. Because water pushes it down

(iv) A student bends a wire but after a while it does not return to its original shape and size. What property of the wire is shown?

A. Plasticity C. Rigidity

B. Elasticity D. Strength

(v) Halima wanted to measure the diameter of a wire. Which of the following instruments should Halima use to do so?

A. A metre rule C. Vernier caliper

B. A tape measure D. A micrometre screw gauge

(vi) Is defined as a big idea or explanation that scientists come up with to describe how something in nature behaves.

A. Law B. Theory C. Principle D. Rule

(vii) Hydrometer is able to float up right on a liquid because:

A. Has a narrow stem C. It contains lead ballast

B. Has a wide bulb D. It has a long stem

(viii) If the relative density of a substance is 11.3, the density of the substance is,

A. 11.3 kg/m3 B. 11.3 g/cm3 C. 11.3 kg/cm3 D. 11.3 g/m3

(ix) Is a force produced on a solid object when it is twisted?

A. Tensional force B. torsional force C. Normal force D. Upthrust

(x) If the radius of a capillary tube is increased while keeping all other factors constant, how will it affect the capillary rise height?

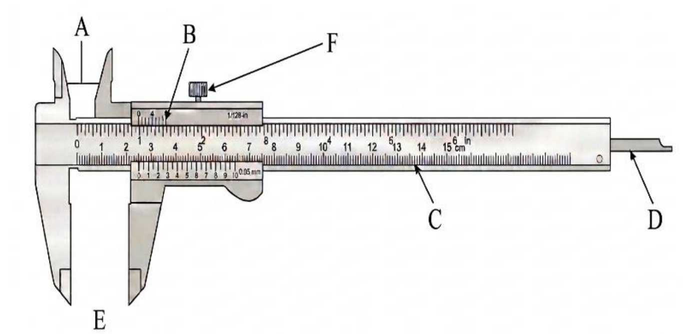
A. The capillary rise will increase

B. The capillary rise will remain the same

C. The capillary rise will decrease

D. The capillary rise will become unpredictable

2. Match the functions of vernier caliper in List A with their corresponding parts in List B by writing the correct answer below the item number in the table provided.

LIST A	LIST B
(i) Is used to measure the internal diameter of a hollow object (ii) Is used to measure the depth of an object. (iii) Is used to secure the objects being measured (iv) Is used to measure the outer diameter of an object (v) It is also called a fixed scale	

SECTION B (70 Marks)

Answer all questions in this section

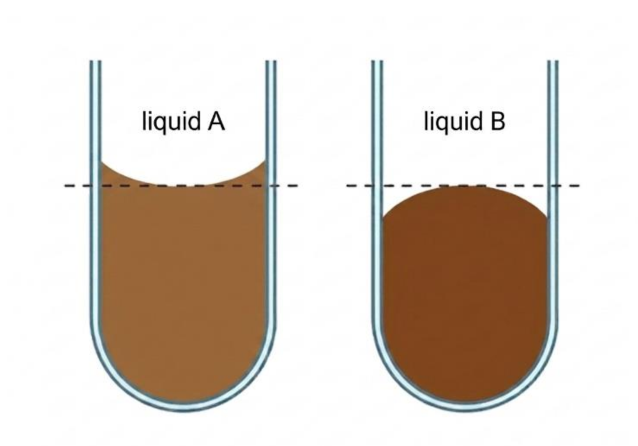
3. (a) A rectangular block measures height 1.00 cm, width 2.50 cm, and length 4.00 cm.

(i) What instrument was used to measure the sides of the rectangular block?

(2 marks)

(ii) Calculate the volume of the rectangular block. **(3 marks)**

(b) A group of students from a certain school were performing an experiment in the physics laboratory, they poured the two liquids into two separate test tubes. They observed that the two liquids had different menisci as shown in the figure below.



(i) Identify the meniscus of each liquid. **(2 marks)**

(ii) Briefly explain why each liquid had its meniscus. **(3 marks)**

4. (a) Explain why a hot soup is tastier than a cold soup? **(5 marks)**

   (b) What is the coefficient of friction, which is calculated in the context of a brick with a mass of 8 kg being pulled on a rough surface with a friction force of 26 N? **(5 marks)**

5. (a) Briefly explain why it takes more time to pour one litre of honey than one litre of water.

   (4 marks)

   (b) If an object has a mass of 200 g on the earth, how much would it weigh on the moon given that the acceleration due to gravity on the moon is one-sixth that of the earth?

   (6 marks)

6. (a) According to Archimedes' principle, when an object is immersed in a fluid, it will experience an upward force called upthrust or buoyant force. Mention three factors affecting the upthrust. **(3 marks)**

   (b) A beaker contains 120 cm^3 of the liquid. A 25 cm^3 pipette is used three times to transfer the liquid to another beaker. Calculate the volume of the liquid left in the original beaker? **(7 marks)**

7. (a) Mr. Ngoma was teaching a lesson on the force of gravity. He asked his students to name its properties, but they all failed to do so. As a Form One student, help your classmates by listing four (4) properties of gravity. **(4 marks)**

(b) A weight of 2.0 N is hung from a spring. The extension produced is 6.0 cm. The weight is removed and a 8.0 N weight is hung from the spring. The spring does not pass its limit of proportionality. What is the new extension of the spring? **(6 marks)**

8. (a) Why is a soap or detergent used in washing clothes? Briefly explain. **(4 marks)**

(b) An iron piece of mass 360 g and a density of 7.8 g/cm^3 is suspended by a rope so that it is partially submerged (halfway) in oil of density 0.9 g/cm^3. Find the tension in the string. **(6 marks)**

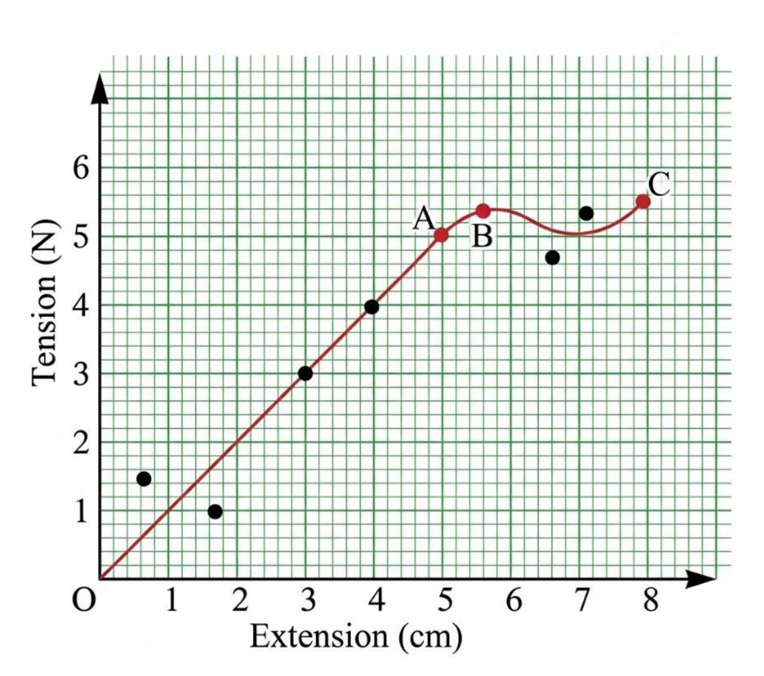
9. (a) Why is it advisable to use a bucket with a thick handle over the one with thin handle? **(4 marks)**

(b) A block of wood weighs 3 N and measures 5 cm x 10 cm x 4 cm. if it is placed on a table with its largest base area or smallest area face in contact with the table, it would exert different pressure on the table. Which surface would exert the largest pressure on the table? **(6 marks)**

SECTION C **(15 Marks)**

Answer question **ten (10).**

10. Study the graph below and then answer the questions that follow



(a) What region shows elastic deformation. **(3 marks)**

(b) State the law which gives the relationship between extension and tension. **(3 marks)**

(c) Identify the point which represents the elastic limit. **(3 marks)**

(d) Define the term elastic limit. **(3 marks)**

(e) What region shows plastic deformation. **(3 marks)**